MIL-P-22561C(YD)
10 May 1982
SUPERSEDING
MIL-P-22561B(YD)
24 September 1976

MILITARY SPECIFICATION

PIPE AND PIPE FITTINGS, ACID RESISTANT

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pipe and pipe fittings for use in acid drainage applications (see 6.5).

1.2 Classification.

1.2.1 Pipe. Acid resistant pipe (straight lengths) shall be of the following compositions and sizes as specified (see 6.2, figure 1 and table I) to be used for above and below grade construction.

Composition A - Glass.
Composition B - Silicon iron.

Form 1 - Mechanical joint.
Form 2 - Bell and Spigot.

1.2.2 Pipe fittings. Acid resistant pipe fittings shall be of the compositions, forms, styles, and sizes as specified (see 6.2).

Composition A - Glass.
Composition B - Silicon iron.

Form 1 - Mechanical joint.
Form 2 - Bell and spigot.

Style A - Elbows. (figure 2, table II).
Style B - Elbow reducers. (figure 3, table III).
Style C - Straight tees. (figure 4, table IV).
Style D - Tee and double tee branches. (figures 5 & 6, table V).
Style E - Y and double Y branches. (figures 7 & 8, table VI).
Style F - Combination Y and 1/8 bend single and double branches. (figures 9 & 10, table VII).
Style G - Double quarter bend. (figure 11, table VIII).
Style H - Reducers. (figure 12, table IX).
Style J - "P", "S", and running traps. (figures 13, 14, & 15, table X).
Style K - Oval and hemispherical cups. (figures 16, 17, & 18, table XI).
Style L - Plugs (caps). (figure 19, table XII).
Style M - Beehive and open and overflows. (figures 20 & 21, table XIII).

FSC 4710

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection for Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A518 - Corrosion-Resistant High Silicon Cast Iron.
E438 - Glasses in Laboratory Apparatus.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Description. This specification establishes the performance and acceptance requirements for acid resistant pipe and pipe fittings. Acid resistant pipe and pipe fittings shall be constructed of the materials specified herein and capable of withstanding the chemical corrosion specified herein.

3.2 Standard commercial product. The pipe and pipe fittings shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the pipe and pipe fittings being furnished. A standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs or brochures and represents the latest production model(s).

* 3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3.1 Composition A. Composition A shall be made of borosilicate glass conforming to ASTM E438.

3.3.2 Composition B. Composition B shall be made of high silicon iron conforming to ASTM A518 and shall be of the form specified.

* 3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

* 3.5 Accessory items. When specified (see 6.2), accessory items such as gaskets, clamps, adapter inserts, traps, traps with cleanout cap plugs, drains, and field cutting kits shall be furnished.

3.6 Chemical resistance. Unless otherwise specified (see 6.2), the pipe and pipe fittings shall be resistant to all acids (except hydrofluoric) at concentrations up to 65 percent and at temperatures up to 100° Fahrenheit. The maximum weight loss under these conditions shall not exceed 0.5 percent per year (see 4.5).

3.7 Other requirements.

3.7.1 Glass. Glass pipe and pipe fittings shall conform to ASTM E438.

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3.7.2 Cast Iron. Cast iron pipe and pipe fittings shall conform to sections 3 through 13 of ASTM A518. These sections include the following options. When required (see 6.2), the options shall be specified in the acquisition document:

a. ASTM A518: Paragraph 6 - Check Analysis.
b. ASTM A518: Paragraph 7 - Transverse Bend Test.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. All pipe and pipe fittings of the same type, composition, form, style, and size offered for delivery at one time shall be considered a lot for the purpose of inspection.

4.3 Sampling. A random sample of the pipe and pipe fittings shall be selected from each lot in accordance with MIL-STD-105.

4.3.1 Sampling for examination. Examination of the pipe and pipe fittings shall be based on inspection level II and an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.3.2 Sampling for tests. Tests of the pipe and pipe fittings shall be based on inspection level S-2 and an AQL of 6.5 percent defective.

4.4 Examination. Each sample selected in accordance with 4.3.1 shall be examined for compliance with the requirements of this specification and the referenced subsidiary specifications to the extent specified herein. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.5 Tests. Each sample selected in accordance with 4.3.2 shall be tested to determine compliance with this specification. Tests shall be conducted as specified in the applicable ASTM standard. Chemical resistance shall be determined in accordance with standard industrial practice and shall conform to the requirements of 3.6.

4.6 Test data. All test data shall be recorded and shall be available to the Government on request.

* 4.7 Packaging inspection. The preservation, packing, and marking shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

* 5.1 Preservation. The preservation shall be level A or commercial as specified (see 6.2).

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5.1.1 Level A.

* 5.1.1.1 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.2 Gaskets. Gaskets shall be preserved and packaged in accordance with method IC-3 of MIL-P-116.

5.1.1.3 Clamps and accessory items. Clamps and accessory items shall be preserved and packaged in accordance with method III of MIL-P-116 in fiberboard boxes conforming to PPP-B-636, class weather-resistant.

5.1.1.4 Pipe and pipe fittings. Pipe and pipe fittings shall be preserved and packaged in accordance with method III of MIL-P-116.

* 5.1.2 Commercial. Material shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Pipe, pipe fittings, gaskets, clamps and accessory items shall be packed in boxes conforming to PPP-B-601, overseas type or PPP-B-621 class 2. The contents shall be cushioned, blocked, and braced to prevent movement and damage.

5.2.2 Level B. Pipe, pipe fittings, gaskets, clamps and accessory items shall be packed as specified for level A except that boxes shall be domestic type or class I as applicable.

* 5.2.3 Commercial. Pipe, pipe fittings, gaskets, clamps and accessory items shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers shall comply with applicable carrier rules and regulations.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Acid resistant pipe and pipe fittings covered by this specification are intended for use above and below grade levels in acid drainage application in laboratories, plating, and battery shops.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and shall include the following information in acquisition documents:

a. Title, number, and date of this specification.
b. Composition, form, and size pipe required (see 1.2.1).
c. Composition, form, style, and size fittings required (see 1.2.2).
d. When specifications and standards are other than as specified (see 2.1.1).

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e. When accessory items are required (see 3.5).
f. When pipe and pipe fittings do not have to resist all acids (see 3.6).
g. When option a. or b. is required (see 3.7.2).
h. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Special requirements. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and samples will not normally be required prior to contract award to determine compliance with this specification. If for any particular purpose, submission of samples in conjunction with bids is considered necessary, they should be specifically asked for in the Invitation for Bids documentation and the particular purpose to be served by the bid sample definitely stated. This specification should apply in all other respects. Installations using these types of pipe and pipe fittings will normally use a combination of compositions, each best suited for that installation. Pipe and pipe fittings required for replacements in existing systems should be specified as to composition, form, and manufacturer due to dimensional differences.

6.5 Specification applicability. This specification does not include all types, designs, forms, and sizes of the commodities indicated by the title of this specification, or which are commercially available. Plastic resin pipe and pipe fittings which have acid resistant properties are included in MIL-P-22245 and other Federal and Military specifications. It is intended to cover the types, compositions, forms, styles, and sizes most commonly used in Government installations.

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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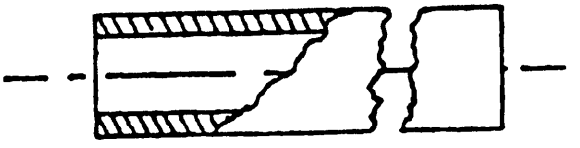


FIGURE 1. Straight pipe.

TABLE I. Straight pipes.

ID Size (inches)	Composition
1-1/2	A & B
2	A & B
3	A & B
4	A & B
6	A & B
8	B
10	B
12	B
15	B

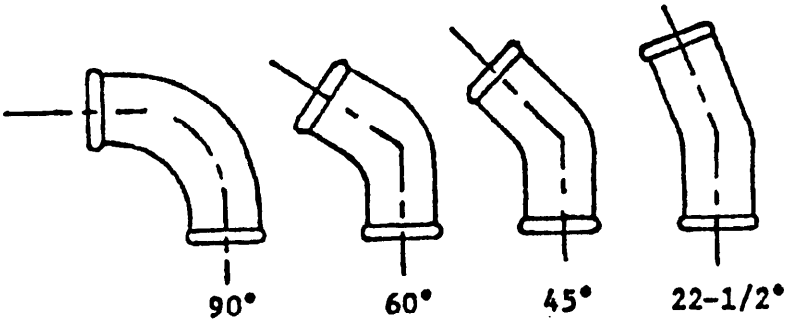


FIGURE 2. Elbows.

TABLE II. Elbows.

ID Size (inches)	Composition			
	90°	60°	45°	22-1/2°
1-1/2	A & B	A & B	A & B	A & B
2	A & B	A & B	A & B	A & B
3	A & B	A & B	A & B	A & B
4	A & B	A & B	A & B	A & B
6	A & B	A & B	A & B	A & B
8	B	B	B	B
10	B		B	
12	B		B	

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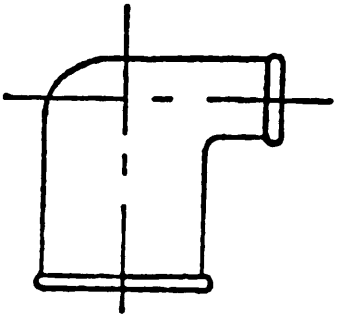


FIGURE 3. Elbow reducer.

TABLE III. Elbow reducers.

ID Size (inches)	Composition
2 x 1-1/2	A & B
3 x 1-1/2	A
3 x 2	A
4 x 1-1/2	A
4 x 2	A
4 x 3	A
6 x 1-1/2	A
6 x 2	A
6 x 3	A
6 x 4	A

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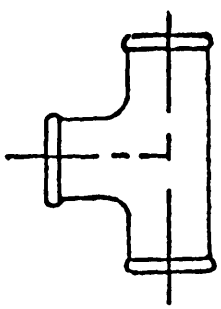


FIGURE 4. Straight tee.

TABLE IV. Straight tees.

ID Size (inches)	Composition
1-1/2 x 1-1/2	A
2 x 1-1/2	A & B
2 x 2	A & B
3 x 2	A & B
3 x 3	A & B
4 x 2	A & B
4 x 3	A & B
4 x 4	A & B
6 x 3	A
6 x 4	A
6 x 6	A

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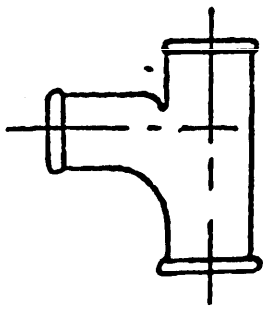


FIGURE 5. Single T branch.

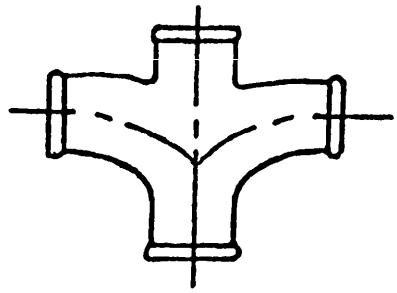


FIGURE 6. Double T branch.

TABLE V. Single and double tee branches.

ID Size (inches)	Composition
1-1/2 x 1-1/2	A & B
1-1/2 x 2	B
2 x 1-1/2	A & B
2 x 2	A & B
3 x 1-1/2	A & B
3 x 2	A & B
3 x 3	A & B
4 x 1-1/2	A & B
4 x 2	A & B
4 x 3	A & B
4 x 4	A & B
6 x 2	A & B
6 x 3	A & B
6 x 4	A & B
6 x 6	A & B
8 x 4	B
8 x 6	B
8 x 8	B
10 x 6	B
10 x 10	B
12 x 8	B

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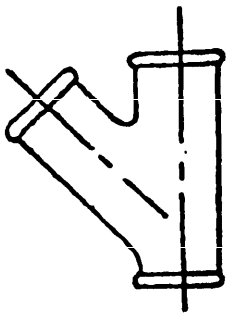


FIGURE 7. Y branch.

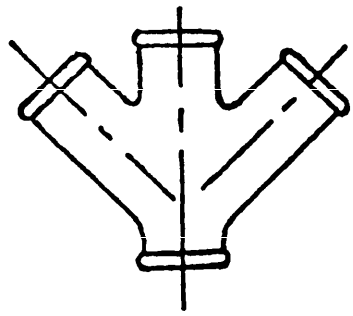


FIGURE 8. Double Y branch.

TABLE VI. Y branches and double Y branches.

ID Size (inches)	Composition
1-1/2 x 1-1/2	A & B
2 x 1-1/2	A & B
2 x 2	A & B
3 x 1-1/2	A & B
3 x 2	A & B
3 x 3	A & B
4 x 1-1/2	A & B
4 x 2	A & B
4 x 3	A & B
4 x 4	A & B
6 x 2	A & B
6 x 3	A & B
6 x 4	A & B
6 x 6	A & B
8 x 2	B
8 x 3	B
8 x 4	B
8 x 6	B
8 x 8	B
10 x 4	B
10 x 10	B
12 x 4	B
12 x 6	B
12 x 8	B
12 x 10	B
12 x 12	B
12 x 15	B

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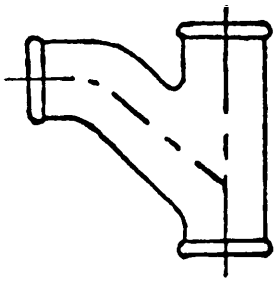


FIGURE 9. Y combination and 1/8 bend, single.

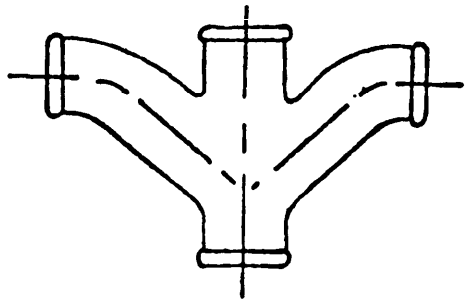


FIGURE 10. Y combination and 1/8 bend, double.

TABLE VII. Combination Y and 1/8 bend single and double branches.

ID Size (inches)	Composition
1-1/2 x 1-1/2	A & B
2 x 1-1/2	A & B
2 x 2	A & B
3 x 1-1/2	A & B
3 x 2	A & B
3 x 3	A & B
4 x 1-1/2	A & B
4 x 2	A & B
4 x 3	A & B
4 x 4	A & B
6 x 2	A & B*
6 x 3	A & B
6 x 4	A & B
6 x 6	A & B
10 x 8	B*

*Single only

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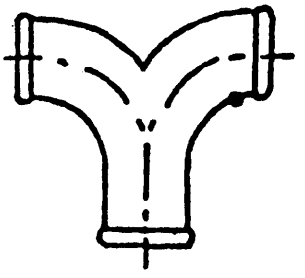


FIGURE 11. Double quarter bend.

TABLE VIII. Double quarter bends.

ID Size (inches)	Composition
1-1/2 x 1-1/2	A & B
2 x 2	A
3 x 3	A
4 x 4	A
6 x 6	A



FIGURE 12. Reducer.

TABLE IX. Reducers.

ID Size (inches)	Composition
2 x 1-1/2	A & B
3 x 1-1/2	A & B
3 x 2	A & B
4 x 1-1/2	A & B
4 x 2	A & B
4 x 3	A & B
6 x 1-1/2	A
6 x 2	A & B
6 x 3	A & B
6 x 4	A & B
8 x 4	B
8 x 6	B
10 x 6	B
10 x 8	B
12 x 6	B
12 x 8	B
12 x 10	B
15 x 6	B
15 x 12	B

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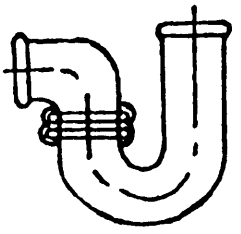
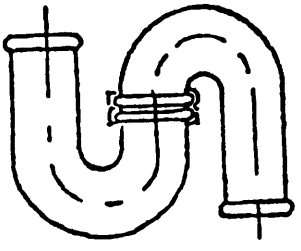
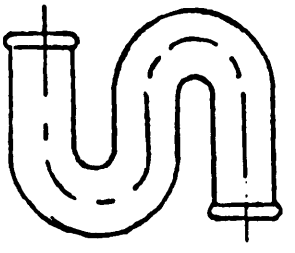


FIGURE 13. "P" trap.



"S" Trap.

Composition A



Composition B

FIGURE 14.

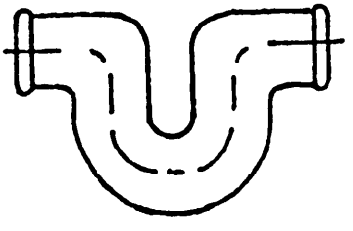


FIGURE 15. Running trap.

TABLE X. "P", "S", and running traps.

Inside diameter size (inches)	Composition
1-1/2 x 1-1/2	A and B
2 x 2-1/2	A and B
2 x 2	A and B
3 x 3	B
4 x 4	B
6 x 6	B
8 x 8	B*
10 x 10	B*

*Running only

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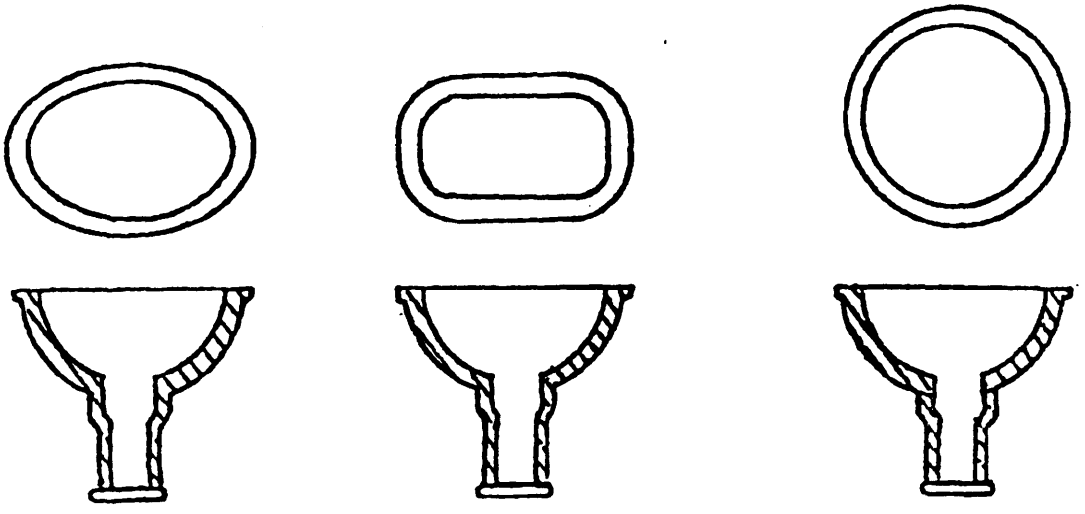


FIGURE 16. Oval Cup. FIGURE 17. Oval cup. (Straight sides) FIGURE 18. Hemispherical cup.

TABLE XI. Oval cups and hemispherical cup.

Size (inches)	Composition
Oval Cup	
3 x 6	A & B
Oval Cup (Straight sides)	
3 x 6	A & B
Hemispherical Cup	
3	A & B
5	B
6	A & B

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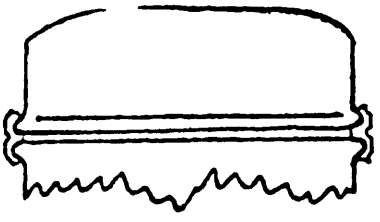


FIGURE 19. Cap plug.

TABLE XII. Plugs (cap).

Size (inches)	Composition A only
1-1/2	A
2	A
3	A
4	A
6	A
8	A
12	A

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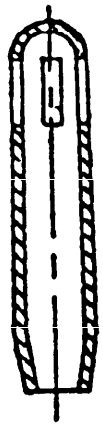


FIGURE 20. Beehive overflow.

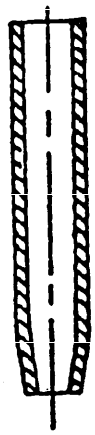


FIGURE 21. Open end overflow.

TABLE XIII. Beehive and open end overflow.

Size (inches)	Length (inches)	Composition**
1	6-1/8*	B
1	10-1/8*	B
1-1/2	2-1/8*	B
1-1/2	4-1/8*	B
1-1/2	5-1/4	A
1-1/2	6-1/8	B
1-1/2	7-1/4	A
1-1/2	8	B
1-1/2	8-1/8	B
1-1/2	9-1/4	A
1-1/2	10	B
1-1/2	10-1/8	B
1-1/2	11-1/4	A
1-1/2	14	B
2	2-1/8*	B
2	4-1/8*	B
2	4-3/4	A
2	5	A
2	6-1/8	B
2	7	A
2	8	B
2	8-1/8	B
2	9	B
2	10	A
2	10-1/8	B
2	11	A
2	14	B
3	6-3/4	A
3	8-3/4	A
3	10-3/4	A

* Beehive only.

** Some manufacturers offer overflows composed of acid-resistant polytetrafluoroethylene (PTFE) compounds which are suitable for the intended use.

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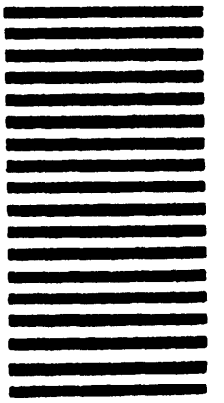
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